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Do I Value My Private Data? Exploring Model-Driven Predictions on the Willingness to Use Smart Home Devices

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ABSTRACT

Smart home devices, increasingly prevalent in modern households, collect large amounts of data from their immediate environment. These data are often of significant monetary value to companies, sometimes without necessarily improving their functionality. As such, the type and scope of user data devices collect may entail privacy and security risks of relevance to the decision which of otherwise similar products to purchase. Our goal is to determine if users consider the value of the data to be collected when planning to use such devices. Drawing on theories from media and cognitive psychology, we proposed two causal and two extended models of privacy behavior and conducted a study to compare the evidence for one over the other respectively. We employed a factorial within-between-subject design ($N_A = 517; N_B = 420$), considering five types of smart home devices, three levels of endogenous data value (severity of risks), and three levels of exogenous data value (likelihood of risks). Bayes Factor model comparisons of the fit multilevel models indicated extreme relative evidence for the model assuming value-dependent privacy behavior over one predicting deviations from rational data valuation. We discuss implications for digital sovereignty and privacy interventions in smart home usage.

Introduction

Privacy in Smart Homes

Smart home devices are no longer fantasies from science fiction novels, but have arrived in millions of modern consumer households (Lasquety-Reyes, 2021), and their popularity is even forecast to further accelerate worldwide in coming years (IDC, 2022). Most users are unaware of the continuous and exhaustive amounts of data these devices collect to enhance convenience (Chalhoub et al., 2020), or even for other purposes not immediately relevant to users, such as product development and marketing. For example, voice-activated assistants

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often possess a large range of abilities, such as posting social media content and providing weather reports, for which they require disclosed audio information, users' login IDs or passwords, and current location data.

Privacy risks associated with smart homes primarily stem from unauthorized access to or loss of data, or compromised devices. Enhancing security and privacy fundamentally rests on technical issues (e.g., encryption methods) as well as on psychological challenges (e.g., motivation) (Aytes et al., 2003; Papp et al., 2019). Researchers in this interdisciplinary field have engaged in developing various conceptual frameworks that describe privacy-related behaviors of users, such as maintenance and updating habits, and their influencing factors.

Across the different conceptual frameworks, the literature paints a picture of two recurring theoretical assumptions: one focuses on more rational decision-making and calculated privacy behavior (which we refer to as assumptions of rationality; A_{ratio}), whereas the other emphasizes certain deviations from rational privacy behavior (described as assumptions of deviations; A_{dev}). This core differentiation is relevant to practical implications supporting the digital sovereignty of smart home users. Assumptions of rationality suggest that users have, e.g., an accurate understanding of risk and benefit boundaries when disclosing information. Privacy risks could then be reduced with training efforts directed toward increased user education and awareness. Assumptions of deviations on the other hand, point to biased privacy perceptions of users that reflect underlying heuristics (Gilovich et al., 2002). To promote digital sovereignty, which is deemed to be severely limited due to discrepancies with rational objectivity, advocates could implement different types of training focusing on other processes, such as task characteristics.

Assumptions of Rationality (A_{ratio})

The assumptions of rationality approach predicts privacy behavior as a function of dynamic privacy processes, e.g., the Privacy Process Model (PPM) by Dienlin (2015). The PPM proposes that each situation has an objectively assessable privacy context with four independent dimensions: the amount of information, number of people having access or being present, intimacy of information, and physical proximity of other people. Individual perceptions of the privacy context may differ from the objective context and can vary across people. In contexts with low perceived privacy, people refrain from disclosing data, whereas in contexts with high perceived privacy, the level of information provision is higher. The perceived privacy and the privacy behavior are compared to the respective desired status of privacy (cf. Altman, 1975) and regulation on controllable context or behavior should occur in cases of discrepancies. The explicitness of context-dependent behavior can also be found in Masur's Theory of

Situation Privacy and Self-Disclosure (2018). Masur's distinction between contexts and situations further emphasizes the influence of psychological states and trait characteristics on individuals' decision-making. Whereas Nissenbaum's Theory of Contextual Integrity (2010) addresses a more sociological perspective. Here, privacy behavior is posited to be primarily shaped by norms, goals, and purposes of data exchanges. This is mirrored in prior established theories, such as the Communication Privacy Management (CPM) Theory (Petronio, 2002; Petronio & Durham, 2008). These authors suggest similar arguments but pinpoint distinct rules that are used to control individual information disclosure. As one critical aspect of privacy rules, CPM theory explicitly factors in contextual risk-benefit ratios.

In Privacy Calculus Theory (Culnan & Armstrong, 1999; Dinev & Hart, 2006), rational decisions on disclosure of information are made based on the individual's anticipation and weighing of benefits and risks. This benefit-risk analysis focuses on perceptions and predicts self-disclosure once benefits exceed the risks. According to Protection Motivation Theory (Rogers, 1975, 1983), the analysis extents to the response behavior of a perceived risk, including its efficacies and reward perceptions of behaviors that might even increase privacy risks. Risks are considered particularly important according to the Disclosure Decision Model (DDM) by Omarzu (2000). The DDM provides rationales for conditions to be met in order to foster self-disclosure, but also describes what information people would then provide and in the case for increasing risks, "more cautious disclosure behavior" (Omarzu, 2000, p. 180) is expected. Other conditions suggested to encourage self-disclosure, such as anonymity (see Hyperpersonal Communication Theory; Walther, 1996), are highlighted by the Internet-Attribute-Perception Model (Schouten et al., 2007) as being influential when viewed as relevant by users.

Overall, the idea of consciously giving up one's privacy and attentively controlling the disclosure of information as part of context-specific decisions was inspired by frameworks like Goffman's (1965) and demonstrates the long tradition of deliberate privacy behavior theories.

Assumptions of Deviations (A_{dev})

In contrast to rational privacy behavior, assumptions of deviations predict heuristic and systematically biased disclosure. Environmental cues, such as uncertainty of risks associated with data loss or disclosure, prompt the adoption of cognitive heuristics (Kahneman et al., 1982; Tversky & Kahneman, 1974). Although heuristics are not irrational in their nature, they can lead to systematic errors, i.e., biases. Biased privacy behaviors would be interpreted as signatures of underlying heuristics (Gilovich et al., 2002).

For example, compression heuristics, an overestimation of smaller risks and underestimation of larger risks (Fischhoff, 2011), could lead to equalization of objectively different privacy contexts. Or affect heuristics, a reliance on valenced affective states as informational sources for risk analysis (Finucane et al., 2000; Slovic et al., 2004), could inverse the relationship between risk and benefit judgment (Skagerlund et al., 2020). This may result in an undervaluation of high risks when faced with increasing benefits of self-disclosure. An undervaluation of future benefits (smaller risks due to higher privacy efforts) over present benefits (using defaults during a device setup to reduce privacy efforts), i.e., a present bias, might contribute to floor-effects of subsequent preferable privacy conditions (Benhabib et al., 2010; O'Donoghue & Rabin, 2015). In other words, individuals might heavily discount future, uncertain privacy risks (potentially even to a degree where privacy contexts are once more equalized) to favor immediate gains that reflect time-inconsistent preferences (O'Donoghue & Rabin, 1998).

Dual-process theories of decision-making also stress the importance of heuristics (default responses) when coping with the many demands of the environment requiring working memory capacity (Evans & Stanovich, 2013; Slovic et al., 2005). In environments where privacy behavior is usually perceived as a secondary task (whereas the primary task in smart homes will often be need gratifications) inhibiting heuristics to favor rational deliberation could be theorized as counterproductive (Kahneman & Tversky, 1973; Pashler, 1998). Specifically regarding this last point, the Limited Capacity Model of Motivated Mediated Message Processing by Lang (1995) asserts that users might not allocate resources to encode information on self-disclosure due to the perceived lack of importance in that moment of decision-making (it being a secondary task). The complexity as well as obscurity of privacy dimensions would require immense resources in the subprocess of information retrieval that are avoidable when evaluation of privacy contexts is not prioritized.

Following arguments that emerge from A_{dev} theories, it is suggested that task characteristics and limited cognitive capacities elicit biased deviations from deliberate privacy behaviors, with self-disclosure being independent from context-specific privacy risks.

The Role of Monetary Data Values

Measuring Data Value

The outlined theories of A_{ratio} and A_{dev} have profoundly impacted their respective research fields. In this work, we aim to introduce a complementary perspective on privacy risks in smart homes related to unauthorized access to or loss of personal data: that is, privacy risks vary depending on the objectively measurable components of recorded data, e.g., the degree of detail or broadcasting, that have a specific price tag.

There are (a) endogenous and (b) exogenous components to the value of data sets. Endogenous components reflect properties of the data set itself, for example the degree of granularity and sensitivity of the data (Kannan et al., 2018; Robinson, 2017). Granular data is detailed or instance-level information (e.g., itemized call lists) and not aggregated or summarized (e.g., total calls), whereas sensitive data might include personal identifiable information (e.g., name and address), health or financial data, or include confidential information (e.g., legal documents). These value components of data sets reflect the private information capital that users own, which Zuboff (2019) discusses economically in terms of raw materials that are provided by users to companies that collect data. They contribute mainly to the severity of risks, which we define as the extent of potential negative impact that an unauthorized access or loss could cause. Given the relationship between value and risk concepts, we will refer to this component as severity of risks value (SEV_{val}). Exogenous components reflect the presence of context, such as the number of people with access to the data. From narrow- to broadcasting (sharing data with a specified small group of people or with loosely defined audiences), generated data sets can be made accessible to different privacy spheres (Acquisti et al., 2013; Danezis et al., 2005). These value components originate from users taxing broader spheres more for accessibility than smaller spheres. Since publicly shared information has a higher likelihood of risks, defined as the probability that risks will occur (e.g., via reanalysis), this is another respective risk level associated with data value: the likelihood of risks value (LH_{val}).

Since a transparent data marketplace does not exist yet, users can assess the value of their data via two strategies: (a) one could consult actual black market prices on the dark web for illegally obtained comparable data sets (Robinson, 2017) or (b) users can assess feasible set properties (i.e., granularity and sensitivity) to at least evaluate relative data values (Kannan et al., 2018). Both strategies highlight inherent uncertainties and resources required for data valuation, while also providing methods to objectively assess privacy contexts.

Our Research Scope

On that account, the pending research question is: Do people actually consider the value of the data collected by smart home devices when they intend to use them? The main goal of this research is to provide causal evidence of the influence of data values on the willingness to use a smart home device. We investigate specifically this variable as different data sets entail a certain monetary worth as well as privacy and security risks. Willingness to use a device represents just one specific manifestation of privacy behavior, even though there are naturally many other distinct manifestations. By focusing on an early-stage privacy scenario, our study does not encompass the full diversity of privacy behaviors or changes over time with increased device familiarity.

Empirical evidence of subjective valuation of privacy has so far largely relied on a popular economic paradigm that measures data value with users' willingness to pay for privacy and willingness to accept privacy loss. Despite other consumer contexts where willingness to pay and willingness to accept lead to valid results, this may not be the case for estimating subjective data worth (Winegar & Sunstein, 2019). Winegar and Sunstein (2019) argue that the main reason is the severely high willingness to pay and willingness to accept ratio found (i.e., users are willing to spend far less to protect data privacy than they would demand to allow access to personal data), which could be explained by (a) the lack of information available to users to be able to estimate the actual value; or (b) morally-led response-giving that reflects the perceived unfairness of paying for data protection that should by norm be protected. A behavioral measure is proposed to be the more relevant operationalization.

In addition, the literature does not seem to clearly favor either one of the outlined theoretical assumptions (A_{ratio} and A_{dev}). It follows that the utility of the assumptions is yet to be explored in the context of data valuation. Hence, we first modeled the competing aspects between A_{ratio} and A_{dev} that at the same time reflect their commonality and then conducted an experimental study about whether data values influence the willingness to use a device. This was to causally test which of these two main models better accounts for the data at hand. We utilized Bayes Factors (BF) to compare posterior distributions of the two models, while varying the prior probability of each model to provide insights into data persuasiveness for A_{ratio}-, A_{dev}-, and neutral study observers.

Modeling Assumptions of Rationality: Effects of Data Value on Privacy Behavior

We aim to provide model predictions that mirror the dichotomous nature of the assumptions, i.e., a point of contrast between rational data valuation and deviating behaviors from the true data value. Our research question is then addressed by examining which of the models provides a better explanation by considering data values (SEV_{val} and LH_{val}) as inversely linear predictors of willingness to use a smart home device. Causal understanding of the valuation process is crucial for future research and consumer protection initiatives, as it can guide the development, testing of effectiveness, and facilitation of interventions (cf. Watts et al., 2018).

Privacy theories that are unified by A_{ratio} suggest users deliberately compromise their privacy and rationally manage the sharing of information as part of decisions tailored to specific contexts (including inherent risks and benefits). As an implication, user-centered IT security research can focus on security behaviors that rely on accurate perceptions of individual and situational risk-benefit boundaries. In our understanding, it is therefore the predictable coherence between data values and resulting privacy behavior that is the determining commonality across this perspective. This implies: *increasing* *data values as conditions of use are expected to lead to a decreasing willingness to agree to use a smart home device.*

Accordingly, we propose the following formulation of A_{ratio} in Model 1. Here, $Agree_{use}$ represents the willingness of people to use a smart home device. SEV_{val} and LH_{val} refer to the severity and likelihood of risks value respectively. Both are data value variables and both include facets of privacy (concealing personal data and limiting members of spheres) as well as associated risks (identification and publicity). We account for variability at the individual and smart home device level, also considering our study employs a within-between-subjects design.

$$Agree_{use} \sim SEV_{val} + LH_{val} + (1|ID) + (1|Device) \tag{1}$$

In addition to this causal and focused exploration of the effect of data value on privacy behavior, we propose a secondary and exploratory model that reflects a broader perspective derived from A_{ratio} theories. Model 2 acknowledges the embedded nature of data value within a network of other predictors, which however are not causally tested in this work.

Perceived benefits are linked to higher self-disclosure (Cheung et al., 2015; Dienlin & Metzger, 2016; Meier & Krämer, 2024b). Here, sharing information might align with an increased agreement to use a device, while privacy concerns increase protection behaviors (H. Li et al., 2011) and reduce the use of the service (Chellappa & Sin, 2005; Meier et al., 2023). This might negatively affect users' willingness to use devices with mandatory data processing (as is the case in this study). Having prior negative privacy experiences (either personal or heard of others') is posited to lead to concerns (Koohikamali et al., 2017), consequently indicating a negative effect on system use as well. General online privacy literacy is also associated with specific data protection measures (Masur et al., 2017) and privacy regulation strategies (Park, 2013) that might cause disagreement of device use. Users' perception of the specific privacy context can be differentiated on a vertical (institutional) and horizontal (social) level (Ayalon & Toch, 2019) by focusing on, e.g., disclosure control. The assumption is that perceived control increases self-disclosure (Brandimarte et al., 2013) and hence, users' willingness to use a device despite high value data processing. In general, the ability to implement measures to protect the disclosed information or other aspects of privacy during system use might positively affect users willingness to use the device (Sarikakis & Winter, 2017).

$$\begin{aligned} Agree_{use} \sim\ & SEV_{val} + LH_{val} + Benefits + Concerns + Experiences_{neg} + Literacy \\ & + Privacy_{vert} + Privacy_{horiz} + Protection_{msr} + (1|ID) + (1|Device) \end{aligned} \tag{2}$$

Modeling Assumptions of Deviations: Effects of Data Value on Privacy Behavior

Theories clustered in A_{dev} do not assume privacy behaviors to be generally unpredictable, but allow a shared prediction that users' willingness to use a smart home device will show some deviations from the true data value. Structural biases are posited to cause an equalization of data value contexts and hence, data value may not predict privacy behavior. As a consequence, privacy researchers should not only be interested in studying the perception of data values, but rather investigate other processes that foster behavioral change of smart home users. It is the discrepancy with an objective data valuation that is common to all included approaches of A_{dev}. To clarify the overarching inference: *data value variables are not linear predictors of the willingness to agree to use a smart home device.*

Therefore, we propose Model 3 as a formulation of A_{dev}. Here, 1 represents the constant. In other words, both the severity and likelihood of risks value as conditions of use have no effect on $Agree_{use}$, the willingness of people to use a smart home device. This approach is based on understanding biases as cues of cognitive heuristics and operationalizing them as behavioral deviations from the true objective data value. Intercepts are again nested in persons and devices for reasons of the variables' variability and the non-independent data structure of our study design. Most importantly, it is the potentially meaningful variation in $Agree_{use}$ of the intercept-only model that we think is appropriately reflecting these shared assumptions across A_{dev}.

$$Agree_{use} \sim 1 + (1|ID) + (1|Device) \tag{3}$$

While our primary objective is to examine data value as a causal influencing factor, we also see the value in exploring another model incorporating other predictors from A_{dev}: Model 4. Both extended models are cross-sectional and of exploratory nature.

Besides equalizing different data values, biases may also lead to other systematic behavioral deviations. An affect bias can lead to affect congruent effects on privacy risk (H. Li et al., 2011) or modify benefit perceptions (Kehr et al., 2013) and hence, might congruently influence the likelihood of agreeing to use a device in this study. Higher familiarity with smart home devices could result in a biased agreement by compensating privacy concerns (Y. Y. Li, 2014; Nepomuceno et al., 2014). Similarly, herding biases might cause a disregard of individually perceived risks (Gerber et al., 2017) and hence, users might make agreement decisions based on the imitation of privacy behavior norms of other users (Auxier et al., 2019; Masur et al., 2021). Lastly, expertise may also encourage users' willingness to adopt a smart home device, perhaps due to the idea of a post-privacy age (Dienlin & Breuer, 2023).

$$\begin{aligned} Agree_{use} \sim & Affect_{pos} + Affect_{neg} + Familiarity + Herding + Expertise \\ & + (1|ID) + (1|Device) \end{aligned} \tag{4}$$

In conclusion, we causally address the research question of whether individuals take into account the value of data collected by smart home devices when intending to use them. The literature presents two competing theoretical assumptions: A_{ratio} and A_{dev}. One main and one secondary, extended model for each perspective was deducted. We now report how we conducted an experimental study to determine which of the main and extended models respectively better explained the observed data.

Materials and Methods

All materials, along with the processing and analysis codes of Stages 1 and 2, are publicly available on OSF.[1] The approved Stage 1 protocol of this registered report was additionally preregistered on OSF prior to data collection.[2]

Design Overview

The study was an online experiment and carried out via Prolific with an incompletely balanced within-between-subjects experimental design: 5 (A: smart home device) × 3 (B: severity of risks value) × 3 (C: likelihood of risks value) factors. Participants were randomly shown product descriptions of three out of five smart home devices (factor A): smart mirror, smart video doorbell, smart room lighting, smart security camera, and smart trash bin. The product descriptions informed the reader about the severity of risks value (factor B), i.e., the data collected by the device, which were represented by three levels: low, medium, or high severity (from less valuable to more valuable data). Further, the product descriptions defined the likelihood of risks value (factor C), i.e., the number of people potentially getting access to the data, again represented by three levels: low, medium, to high likelihood (from highly restricted access to less restricted access). For each product, both the level of severity and likelihood of risks value were randomly combined. Factor levels for factors B and C were drawn with replacement; however, factor-level combinations of B and C were drawn without replacement (with the exception of a 0.04% chance of a doublet combination due to limitations of the Qualtrics platform). As such, out of $B : 3 \text{ x } C : 3 = 9$ possible combinations, each subject saw only three in total. The main dependent variable was the participants' indicated willingness to try out the device at home.

Main Independent Factors

We piloted materials of the two main independent factors in two pre-studies with $N_{pre1} = 48$ and $N_{pre2} = 49$. A detailed description of the pre-studies and their findings is provided on OSF.[3] The exact wording of each level is also provided on OSF.[4]

The three levels of SEV_{val} were created by differentiating degrees of data set granularity and sensitivity: (1) low value data contains only summary information about traffic data (also called communications data) without sensitive information being stored; (2) medium value data renders detailed information about traffic data without sensitive information being stored; (3) high value data includes traffic as well as content data on an instance-level and some sensitive variables (e.g., health data) that are stored.

The LH_{val} factor levels capture audiences of different size (potentially) accessing the generated data set: (1) data that is stored and accessed by the research group has a low likelihood of risks value; (2) data that is stored and accessed by third parties located in the EU (e.g., companies involved with the product) has a medium likelihood of risks value; (3) data that is stored and accessible via a public research repository has a high likelihood of risks value.

Sample

Data collection started after the study's preregistration on OSF[5] and closed when the intended sample size was reached. Given the current scarcity of practical guidance available for designing research studies that utilize BFs for a posterior model comparison as well as the lack of information regarding posterior probability distributions of parameter values from previous studies about the relationship between monetary data value and valuation behavior, we resorted to a fixed-N design with a sample size of $N = 500$. This sample size also reflects resource efficiency and restrictions (Lakens, 2022). We surveyed a total of 550 participants, accounting for unfinished or not eligible data.

This study sample had a median age of 38 years (ranging between 18–82), with 297 (54.0%) female, 250 (45.5%) male, 2 (0.4%) non-binary participants, and 1 (0.2%) preferring not to disclose their gender. A total of 427 (77.6%) participants owned a smart home device, 327 (59.5%) were full-time employees, and 383 (69.6%) had a Bachelor's degree or higher. For a detailed demographic description of several sub-samples see Table A1 in Appendix A.

The sample was recruited through Prolific and had to meet three participation requirements: (a) a minimum age of 18 years, (b) the first language had to be English, and (c) in a Prolific filter study, participants must have indicated their general willingness to participate in product tests and usability studies.[6] Having recorded participants' general willingness allows us to attribute decisions made in the main study to our manipulation rather than a general attitude toward any usability study.

Procedure

The study was approved by the ethics committee of the Faculty of Psychology at Ruhr University Bochum (reference number: 779).[7] At the beginning, participants provided informed consent and filled in their personal Prolific

ID. Next, there were three trials of value scenarios. Each trial started with a picture and description of a smart home device (randomly chosen from a pool of five devices). The product description included the degrees of data set granularity and sensitivity collected by the device (SEV_{val}) as well as a description of the differently sized audiences with access to the generated data set (LH_{val}); each randomly chosen from one of three levels. All written descriptions were coupled with an item that asked whether the subject has read the text. At the end of each trial, participants were required to indicate their willingness to try out the respective device at home in a usability study. The item responses were displayed in a random order. After three trials, participants were asked to answer several self-report questionnaires and to correctly order the levels of both factors (low SEV_{val} or LH_{val} to high SEV_{val} or LH_{val}). A description of the rules that ought to be applied to the order was provided as instructions to the task. The ordering tasks were presented in random sequence. Trials 1 and 3 further included an attention check. At the end, a debriefing was implemented. The study's complete materials can be found on OSF.[8]

Measures

Privacy Behavior

We chose to observe individual data valuation at the first front of privacy decisions: the willingness of a potential user to try out certain smart home devices. Our operationalization mirrors this case of data valuation not by relying on self-reports of specific data values, but by measuring decisions that indicate privacy behavior (see recommendations from Winegar and Sunstein (2019)).

One dichotomous item measured the participants' privacy behavior. The items' wordings were as follows: "Please let us know whether you would [not] be willing to try out this device at home (e.g., as part of a user study on its features)".

Predictor Variables of Secondary Models

We measured perceived benefits with a scale adapted from Meier and Krämer (2024b), general online privacy concerns with an adaption of the privacy concerns instrument developed by Buchanan et al. (2007), and prior negative privacy experiences with a scale adapted from Smith et al. (1996). To assess general online privacy literacy, we adapted the instrument of Meier and Krämer (2024a). For the assessment of both vertical and horizontal privacy perceptions, we used the instrument developed by Ayalon and Toch (2019) and to measure behavioral privacy protection measures, we adapted the scale from Adhikari and Panda (2018). We used a short form of the positive and negative affect schedule to assess affect biases (Mackinnon et al., 1999) and measured familiarity and expertise with one self-developed item each. To assess herding, we developed two items. Tabularized items, sources, response formats, and adaptions are provided on OSF.[9]

Demographic Variables

Items that surveyed demographic variables were only included in our Prolific filter study. Age, gender, education, employment status, and smart home ownership were surveyed. Item wording and response options are provided on OSF.[10]

Analytic Method

Eligible Data

Exclusion criteria at the level of participants were based on (a) participation requirement, (b) data completeness, and (c) data quality. Factor combination and manipulation checks were applied at the level of data within participants for exclusions. Table 1 lists all criteria in more detail and provides the number of exclusions in the order they were applied.

Our analysis included a split of the data so that two parallel sets of results are presented. Data set A (sample incl. incorrect ranking) does not make any exclusions in addition to the exclusion criteria listed in Table 1; rendering $N_{\mathrm{A}} = 517$ participants and $N_{\mathrm{A\_trials}} = 1540$ trials eligible. Data set B (sample with correct ranking) additionally excludes individuals from all analyses who made at least one error in the ordering tasks; resulting in $N_{\mathrm{B}} = 420$ and $N_{\mathrm{B\_trials}} = 1256$.

Main Model Comparison

We proposed two models (1 and 3) as formulations of their respective theoretical assumptions. Model 1 assumes data value to be a probable, inversely linear predictor of privacy behavior, whereas Model 3 infers no effect of data value on said privacy behavior. The objective of the model comparison is to analyze under which model the observed data is more probable.

Before models were compared, data was screened for accuracy and missingness within the dependent variable. The models were fit via Stan using the brms R package (version 2.18.0; Bürkner, 2017) with the default, weakly regularizing priors (to aid model convergence while not imposing strong prior beliefs on a given parameter) and a Bernoulli response distribution with a logit link function. Models were checked for convergence by (a) inspecting effective sample size measures (ESS) and the potential scale reduction factor on split chains (Rhat), (b) conducting visual posterior predictive checks that compare the posterior samples against the observed data, and (c) plotting caterpillar plots of ran chains to look for adequate overlap. Density plots indicated whether the data has overwhelmed our specified priors (increasing data strengthens the data's influence on the posterior). All model convergence checks provided satisfactory results and are described in detail on OSF.[12] We used BFs as a measure of relative evidence for one model over the other. Evidence in favor of one of the models was interpreted according to Jeffreys' rules (Jeffreys, 1961).

We also demonstrated the BF model comparison taking into account assumed prior probabilities between models by three groups of study observers: (a) neutral observers assuming equal prior probability of both models, (b) proponents of A_{ratio} assuming a prior 95% probability of Model 1 over Model 3, and (c) proponents of A_{dev} assuming a prior 5% probability of Model 1 over Model 3. Further, marginal R2 was computed to assess the size of difference in the utility of the models with regard to the explained variance by the fixed factors.

To test the assumption of A_{ratio} about the factor level ordering, we calculated and visually inspected the estimated marginal means (emmeans) of Model 1. This was necessary given that BF results could, e.g., favor a fixed effect model that includes SEV_{val} and LH_{val}, but as a variant with incoherent factor ordering, and thus, representing neither modeled assumption. Differences between smart home devices were explored descriptively by plotting the estimated intercepts of both models separately for each device.

Secondary Model Comparison

The approach used in the main model comparison was also applied to the secondary model comparison of Model 2 and Model 4. The comparison of the secondary models relied on predictors from different levels of causality, including cross-sectional self-report constructs, and was intended as an exploratory analysis. In addition to the analytic methods described in the previous section, we compared the extended models using PSIS-LOO due to the vast difference in complexity. This represents only a first attempt to formalize the overarching assumptions of various theoretical perspectives and their separately posited effects on privacy behavior.

Results

Data, code files, and result figures are available on OSF.[13]

Table 1. Data exclusion criteria.

Level	Criterion	Condition	Exclusion from	# Excl
Participant	Participation requirement	Mismatched Prolific ID[11]	All analyses	0
Participant	Data completeness	Missing data	All analyses	0
Participant	Data quality	$\geq$ 1 failed attention check	All analyses	27
Within participant	Factor combination check	Combination doublet	Analyses of doubled trials	0
Within participant	Manipulation check	Not read independent factor	Analyses of single trial	29

Across the randomly assigned data value factor levels for data set A (sample incl. incorrect ranking), participants expressed willingness to use smart home devices 998 times, whereas unwillingness was expressed 542 times. For data set B (sample with correct ranking), willingness to use smart home devices was indicated 786 times, whereas in 470 cases participants were not willing.

Outcomes of Main Model Comparison

Regardless of which model priors were used, results provided strong evidence in favor of Model 1 (A_{ratio}) over Model 3 (A_{dev}). For the priors representing a theoretically neutral observer, we found $BF_A = 15,421,458$ and $BF_B = 1,325,072$ in favor of Model 1 over Model 3. For the proponent of Model 3, with $BF_A = 811,656$ and $BF_B = 69,741$ results indicated evidence in favor of Model 1 over Model 3. For the proponent of Model 1, we found $BF_A = 293,007,694$ and $BF_B = 25,176,371$ in favor of Model 1 over Model 3.

Table 2 lists the proportion of explained variances by the models. Results indicated descriptively less explained variance by the respective fixed factors and random effects (conditional R^2) for Model 3 compared to Model 1. The same pattern was found for marginal R^2 values regarding explained variance by fixed factors alone, i.e., intercepts and predictor variables.

Figure B1 in Appendix B illustrates the estimated marginal mean probability of agreeing to using a smart home device. Results showed a slight tendency of coherent factor ordering with A_{ratio} assumptions, though posterior density intervals were considerably overlapping.

We also visualized the estimated random intercepts for each smart home device (see Figure C1 in Appendix C), finding that smart lighting had the highest positive shift from the grand mean, while smart cameras and trash devices showed negative deviations, indicating a lower than average probability of usage agreement.

Outcomes of Secondary Model Comparison

Table D1 in Appendix D shows the descriptive statistics of the variables included in the secondary models.

The comparison of our secondary models provided again extreme evidence in favor of Model 2 (A_{ratio}) over Model 4 (A_{dev}). For a neutral observer, with $BF_A = 173,163,168$ and $BF_B = 77,169,423$ results were in favor of Model 2 over Model 4. For the proponent of Model 4, the model comparison yielded $BF_A = 9,113,851$ and $BF_B = 4,061,549$ for Model 2 over Model 4. For the priors reflecting the proponent of Model 2, we found $BF_A = 3,290,100,198$ and $BF_B = 1,466,219,040$ in favor of evidence for Model 2 over Model 4.

To account for the complexity difference between models (i.e., number of included predictors), model comparison was also performed via PSIS-LOO computations that allow penalizing for complexity. Results indicated that Model 2 is expected to have better predictive accuracy (detailed analyses provided on OSF[14]). Model 2 also descriptively demonstrated higher R^2 values compared to Model 4 (see Table 2).

Figure B2 in Appendix B shows a subtle inclination toward theory-consistent factor ordering, with the exception of low and medium levels of SEV_{val}. Estimated random intercepts for each device (see Figure C2 in Appendix C) revealed similar results to the findings within the main models.

Discussion

By examining the causal impact of individuals' valuations of the data collected by smart home devices on their willingness to use such devices, this study contributed to the exploration of the utility of two recurring theoretical assumptions for data valuation contexts: (a) users rationally manage data sharing, i.e., demonstrate context-dependent privacy behavior (A_{ratio} formulized in Model 1) or (b) users' privacy behavior deviates from calculated disclosure and is independent of context-specific risks (A_{dev} formulized in Model 3). Our results provided extreme evidence for the support of A_{ratio} over A_{dev}, even for model priors representing strong proponents of A_{dev}. The observed data is more probable under the assumption that objective data values influence users' willingness to use smart home devices. These results were further supported when exploring a broader network of predictors within each perspective, albeit while sacrificing the ability to make causal claims.

Aligning with our findings, Malheiros et al. (2013) observed for credit card applications a tendency to withhold more sensitive data (SEV_{val}) and Danezis et al. (2005) identified a greater perceived value for location data shared within certain privacy spheres (LH_{val}). Other studies emphasize both, the influence of SEV_{val} and the importance of tailored preferences for privacy context fits (DiMatteo et al., 2018; Nicholas et al., 2019), whereas we found the probability of usage agreement to be descriptively independent of contextual fit (e.g.,

Table 2. Proportion of explained variances by models for data set A and B.

Model	Component	R^2_A	Lower CI_A	Higher CI_A	R^2_B	Lower CI_B	Higher CI_B
Model 1	Conditional	21.89%	16.56%	26.93%	19.50%	13.72%	25.18%
Model 3	Conditional	16.81%	11.43%	21.95%	13.47%	7.45%	19.26%
Model 1	Marginal	4.73%	2.11%	7.65%	5.17%	2.44%	8.27%
Model 3	Marginal	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Model 2	Conditional	22.72%	17.55%	27.64%	20.66%	14.90%	26.13%
Model 4	Conditional	17.35%	12.02%	22.53%	14.30%	8.35%	20.11%
Model 2	Marginal	11.66%	7.26%	15.82%	11.29%	7.36%	15.41%
Model 4	Marginal	5.83%	2.85%	8.84%	4.37%	2.00%	7.18%

health data for smart lighting). The absence of evidence for users equalizing objectively different data values in our study may be due to two factors: we examined privacy behavior before users engaged with products (Ibdah et al., 2021; Kulyk et al., 2018), and we required participants to read short descriptions of data values rather than the lengthy, difficult-to-read privacy policies typically used in practice (see Obar & Oeldorf-Hirsch, 2020, for an extreme example, that is agreeing to sell one's first-born child).

Our contribution extends to the conceptualization of privacy and security risks by introducing a complementary framework that acknowledges how these risks vary based on measurable monetary components of data and builds on related works from economic research fields by measuring valuation decisions that reflect privacy behavior more behaviorally.

Implications

Given that A_{ratio} models provided better explanations of our data by taking data value factors (SEV_{val} and LH_{val}) into account, evidence for context-specific privacy behavior of smart home users is indicated. Even when we included participants in the analyses who incorrectly ordered data values when asked to show their comprehension (data set A), modeled assumptions of rational privacy behavior still found more support.

This means that under specific circumstances users employed objectively accurate privacy behavior regarding risks (information disclosure) and reward (device usage). Another theory that could be interpreted as aligning with this finding is the Resource Exchange Theory (Donnenwerth & Foa, 1974; Foa, 1971). This theory posits that individuals engage in rational exchanges of resources to fulfill their needs, often contributing to social systems where personal resources, such as information, are shared in return for benefits like services or other rewards. The circumstances of exchange in this study might include a relatively short and transparent data value statement made prior to the engagement with the product (cf. Knowles & Conchie, 2023). This implication is supported by Steinfeld (2016) and Chen et al. (2024) who amongst others point to user-centered benefits of privacy policies and labels that are presented by default with high information density. Researchers or practitioners interested in further detailed suggestions might find publications concerning the description of data collection purposes (Kyi et al., 2024) or the design of privacy policies with a one-pager approach (Bahrini et al., 2022) beneficial additions. In terms of digital sovereignty of smart home users, the evidence for value-adherent privacy behavior in our study suggests that it can, in principle, be upheld.

Relating our findings to the widely discussed privacy paradox (cf. Kokolakis, 2017) – which posits that privacy concerns and similar constructed attitudes fail to predict actual privacy behaviors, leading to seemingly paradoxical information disclosure online (cf. Barnes, 2006; Barth & de Jong, 2017) – it could be argued that the extreme evidence favoring A_{ratio} models over A_{dev} models also opposes its main argument. While our previous elaborations of A_{dev} have focused on the systematic equalization of privacy contexts from a more general perspective on human behavior, the privacy paradox instead is specified for the context of privacy and assumes that privacy behavior erratically deviates from what privacy concerns would predict. Nonetheless, this core argument is similarly represented in the formulation of A_{dev}, indicating that both lines of research could be seen as encapsulated within Model 3. This suggests that the findings can also be interpreted as having implications for the privacy paradox, providing evidence against it (for further discussion of the privacy calculus and privacy paradox approaches see, Dienlin & Sun, 2021).

Educational efforts aiming to extend digital sovereignty may be constructed on the premise that users show an accurate valuation behavior with these kinds of data collection policies. We recommend to design interventions that focus on tangible privacy experiences, for example, with the help of virtual environments visualizing disclosure mechanisms. These interventions offer the benefit of being easily disseminated and accessible as at-home solutions. Supported by databases with integrated device-specific privacy policies, they enable users to verify their data valuation and assess how well their privacy preferences are currently met. These recommendations stand in stark contrast to current market practices, including the frequent use of dark patterns in consent scenarios, because the exploitation of data has become integral to the business models of many organizations (Zuboff, 2019). However, policymakers may find it useful to enforce stringently current regulations (Ruschemeier, 2024) or offer label certifications for data handling to further align business incentives with consumer protection.

Our findings also warrant a reevaluation of the conditions under which A_{dev} applies. The partly uncertain centrality of the deviation from rationality assumption within the discussed theories opens the path to: (a) some of the theories might simply adjust or abandon their affected secondary assumptions or (b) if rationality is deemed a more fundamental core assumption in other theories, then altering those cores can lead to paradigm shifts (cf. Lakatos, 1978). This obscured nature of the assumptions now invites discussion and further empirical exploration of anomalies by proponents of each framework.

Limitations and Future Work

The evidence supporting A_{ratio} should be interpreted in light of its potentially limited robustness. A key design limitation of our study is that we focused on specific data set characteristics and did not explore how individuals value the data of others (cf. Hasan et al., 2023). This could be relevant in shared environments like smart homes, where the presence of household members or guests introduces data ownership considerations. Examining data valuation of at-risk populations would provide further insights into the applicability of our results. Such data might include information on sensitive locations, such as healthcare facilities.

Additionally, our study was limited to relative valuation and did not account for the impact of tangible monetary rewards. With business models (e.g., Tapestri) on the horizon that offer transparent payments for data, so called zero-party data, questions of fairness arise. Future research could explore how monetary compensation affects privacy behavior, particularly examining whether lower payment thresholds influence disclosure. Others may also examine different measures of privacy behavior, such as data obfuscation techniques or changes in privacy behavior over time, to better understand how these behaviors evolve. This would provide a more comprehensive understanding of privacy behavior in relation to data valuation.

Our interpretation of this study's outcomes is limited also in terms of external validity to the specific circumstances of the experimental setting, where participants interacted with unfamiliar products and were not exposed to complex privacy policies. Still, similar context-dependent privacy behaviors have been observed in other domains than smart homes, including financial services (Malheiros et al., 2013), location tracking for purposes of research or commercial interest (Danezis et al., 2005), and mental health support (DiMatteo et al., 2018; Nicholas et al., 2019).

On a more general note, when developing theories, it is essential for the research community to aim at formalizing what are currently loose verbal theories to provide testable predictions for empirical research (Fried, 2020). Formalizing verbal assumptions into mathematical descriptions ensures that theory-driven models reflect the original intent of assumptions, offering logically valid predictions. This approach not only enhances the clarity of theoretical assumptions but also facilitates rigorous testing. Given the lack of a standardized approach for translating theoretical assumptions into model equations (cf. Vanpaemel, 2010), establishing clear instantiations is key to advancing theory development.

Another area of limitations concerns the estimated marginal means that only slightly supported the assumed order restriction of A_{ratio}, indicating that privacy behavior may solely shift in response to extreme differences in values. Also, information criteria estimates were unreliable or discordant, potentially pointing to overall weak models. At this point, our formalizations would benefit from being tested against alternative models, allowing us to compare the robustness of these frameworks across mathematical formulations. Future work could also explore the utility of the additional predictors in the secondary Model 2 by causally testing their influence on privacy behaviors. This approach would enable a direct comparison of evidence supporting either our main Model 1 or a more complex A_{ratio} alternative with additional predictors, such as Model 2, evaluated within a shared causal framework. In our analysis, this comparison would de-emphasize the present causal differences between the models.

Conclusion

This study investigated whether users employ the monetary value of data when deciding to use smart home devices. Our findings strongly support that privacy behavior is adjusted based on the data value context, aligning with the A_{ratio} perspective, as opposed to privacy behavior deviating from rational context-specific disclosure. This understanding opens the door for future work on data value disclosure statements. Leveraging these insights in privacy tools can contribute to further advancing the digital sovereignty of smart home users.

Notes

1. OSF link: https://doi.org/10.17605/OSF.IO/UQJ8V OSF project page
2. OSF link: https://doi.org/10.17605/OSF.IO/ec7up OSF preregistration
3. OSF link: https://osf.io/83kgs pre-studies
4. OSF link: https://osf.io/sxuy7 main study materials
5. OSF link: https://doi.org/10.17605/OSF.IO/ec7up OSF preregistration
6. OSF link: https://osf.io/3muqd filter study design and materials
7. OSF link: https://osf.io/gs3cd ethics approval
8. OSF link: https://osf.io/tg7fb complete survey
9. OSF link: https://osf.io/sxuy7 main study materials
10. OSF link: https://osf.io/3muqd filter study design and materials
11. Participants were contacted when their stated Prolific ID did not match the filter study's ID list. In cases that no match could be concluded, the data was excluded from all analyses.
12. OSF link: https://osf.io/we42n main study supporting analyses
13. OSF link: https://doi.org/10.17605/OSF.IO/UQJ8V OSF project page
14. OSF link: https://osf.io/we42n main study supporting analyses

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Data Availability Statement

All data, code, and materials are available on the OSF repository (OSF link: https://doi.org/10.17605/OSF.IO/UQJ8V OSF project page).

Open Scholarship Data

This article has earned the Center for Open Science badges for Open Data, Open Materials and Preregistered. The data and materials are openly accessible at https://doi.org/10.17605/OSF.IO/UQJ8V.

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Appendices

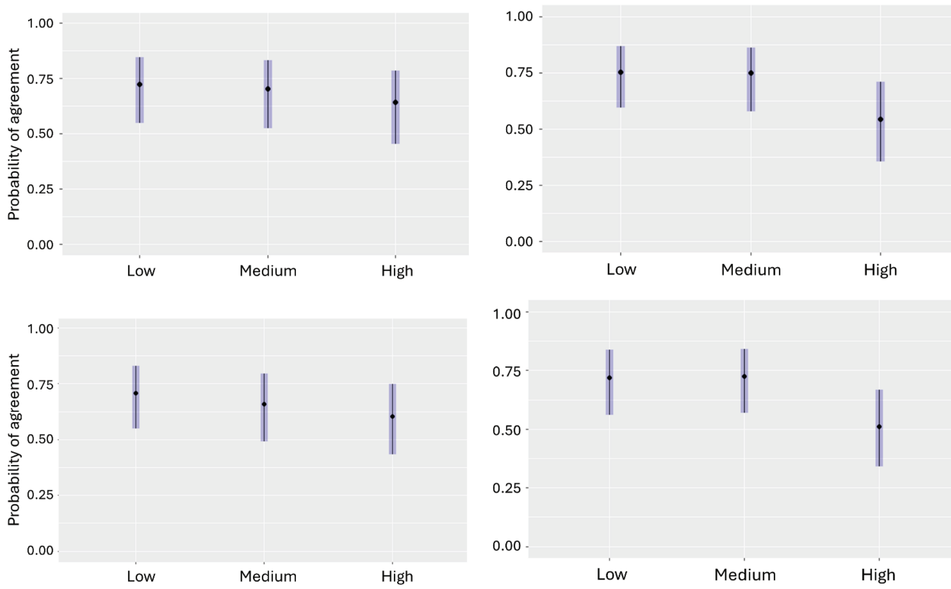
Appendix A. Sample Demographics

Table A1. Descriptive Demographics of (Sub-)Samples.

	Generally No[1]	Generally Yes[1]	Not Participated[2]	Participated[2]	Data Set A[3]	Data Set B[4]	Overall
	($N = 130$)	($N = 825$)	($N = 405$)	($N = 550$)	($N = 517$)	($N = 420$)	($N = 955$)
Age in Years:							
Mean (SD)	39.3 (15.4)	39.0 (12.6)	37.8 (13.3)	39.9 (12.8)	39.8 (12.8)	39.5 (12.7)	39.0 (13.0)
Median [Min, Max]	35 [18, 78]	36.0 [18, 82]	35 [18, 78]	38 [18, 82]	38 [18, 82]	37.5 [18, 82]	36 [18, 82]
Gender:							
Female	74 (56.9%)	446 (54.1%)	223 (55.1%)	297 (54.0%)	276 (53.4%)	230 (54.8%)	520 (54.5%)
Male	53 (40.8%)	375 (45.5%)	178 (44.0%)	250 (45.5%)	238 (46.0%)	187 (44.5%)	428 (44.8%)
Non-binary	2 (1.5%)	3 (0.4%)	3 (0.7%)	2 (0.4%)	2 (0.4%)	2 (0.5%)	5 (0.5%)
Prefer not to say	1 (0.8%)	1 (0.1%)	1 (0.2%)	1 (0.2%)	1 (0.2%)	1 (0.2%)	2 (0.2%)
Smart Home Ownership:							
No	50 (38.5%)	189 (22.9%)	116 (28.6%)	123 (22.4%)	118 (22.8%)	101 (24.0%)	239 (25.0%)
Yes	80 (61.5%)	636 (77.1%)	289 (71.4%)	427 (77.6%)	399 (77.2%)	319 (76.0%)	716 (75.0%)
Employment:							
Full-time	55 (42.3%)	503 (61.0%)	231 (57.0%)	327 (59.5%)	310 (60.0%)	248 (59.0%)	558 (58.4%)
Part-time	24 (18.5%)	126 (15.3%)	58 (14.3%)	92 (16.7%)	85 (16.4%)	69 (16.4%)	150 (15.7%)
Student	11 (8.5%)	45 (5.5%)	29 (7.2%)	27 (4.9%)	25 (4.8%)	23 (5.5%)	56 (5.9%)
Retired	13 (10.0%)	48 (5.8%)	23 (5.7%)	38 (6.9%)	37 (7.2%)	31 (7.4%)	61 (6.4%)
Seeking opport.	14 (10.8%)	38 (4.6%)	28 (6.9%)	24 (4.4%)	22 (4.3%)	16 (3.8%)	52 (5.4%)
Other	9 (6.9%)	58 (7.0%)	31 (7.7%)	36 (6.5%)	32 (6.2%)	29 (6.9%)	67 (7.0%)
Prefer not to say	4 (3.1%)	7 (0.8%)	5 (1.2%)	6 (1.1%)	6 (1.2%)	4 (1.0%)	11 (1.2%)
Education:							
Ph.D. or higher	3 (2.3%)	17 (2.1%)	8 (2.0%)	12 (2.2%)	12 (2.3%)	10 (2.4%)	20 (2.1%)
Master's degree	23 (17.7%)	143 (17.3%)	68 (16.8%)	98 (17.8%)	94 (18.2%)	82 (19.5%)	166 (17.4%)
Bachelor's degree	46 (35.4%)	404 (49.0%)	177 (43.7%)	273 (49.6%)	256 (49.5%)	214 (51.0%)	450 (47.1%)
Trade school	14 (10.8%)	67 (8.1%)	105 (25.9%)	121 (22.0%)	35 (6.8%)	26 (6.2%)	226 (23.7%)
Secondary school	40 (30.8%)	186 (22.5%)	1 (0.2%)	0 (0%)	117 (22.6%)	85 (20.2%)	1 (0.1%)
Primary school	0 (0%)	1 (0.1%)	39 (9.6%)	42 (7.6%)	0 (0%)	0 (0%)	81 (8.5%)
Prefer not to say	4 (3.1%)	7 (0.8%)	7 (1.7%)	4 (0.7%)	3 (0.6%)	3 (0.7%)	11 (1.2%)

[1]Willingness to participate in product tests and usability studies as indicated in the filter study. Only the "Generally Yes" sample was eligible for the main study; [2]Participation in the main study relative to the overall filter study sample; [3]Sample inclusive incorrect solutions to the factor ranking tasks; [4]Sample only with correct solutions to the factor ranking tasks.

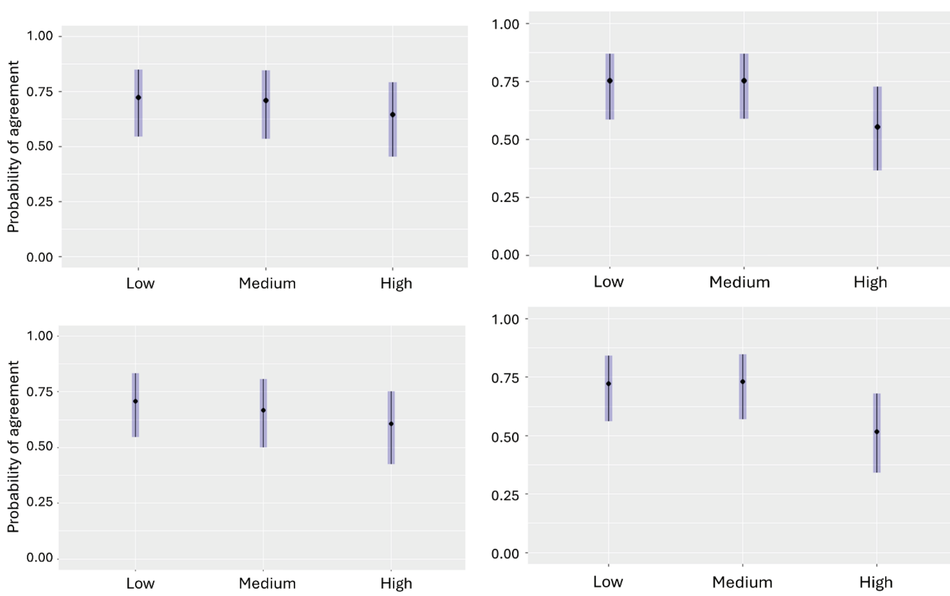
Appendix B. Factor Ordering



(a) Likelihood of risks value

(b) Severity of risks value

Figure B1. Estimated marginal means of willingness to use smart home devices for LH_{val} and SEV_{val} levels in Model 1. Top Row: Results from data set A. Bottom Row: Results from data set B.

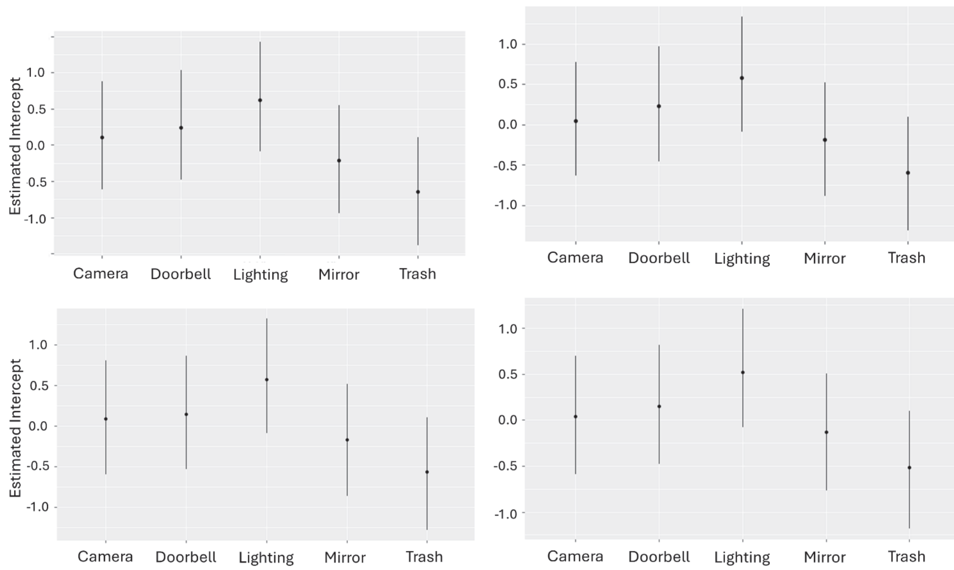


(a) Likelihood of risks value

(b) Severity of risks value

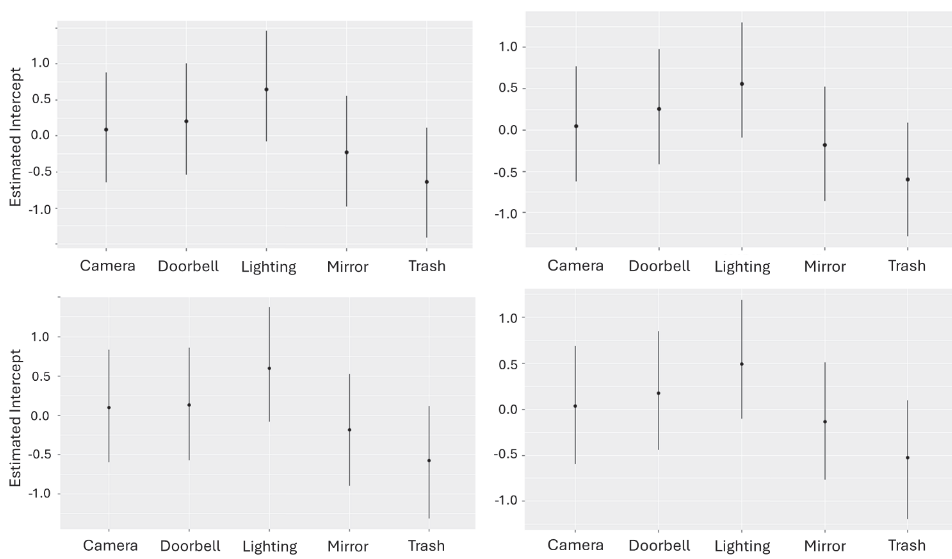
Figure B2. Estimated marginal means of willingness to use smart home devices for LH_{val} and SEV_{val} levels within Model 2. Top Row: Results from data set A. Bottom Row: Results from data set B.

Appendix C. Random Intercepts for Smart Home Devices



(a) Model 1 (b) Model 3

Figure C1. Estimated random intercepts of willingness to use a smart home device for each device respectively within Model 1 and 3. Top Row: Results from data set A. Bottom Row: Results from data set B.



(a) Model 2 (b) Model 4

Figure C2. Estimated random intercepts of willingness to use a smart home device for each device respectively within Model 2 and 4. Top Row: Results from data set A. Bottom Row: Results from data set B.

Appendix D. Descriptive Statistics

Some response distributions of the variables from the extended models seem noticeable skewed, such as low negative affect and privacy perceptions or high levels of herding and expertise. Results also indicate low reliability coefficients (< 0.4) for negative experiences, herding, and literacy.

Table D1. Descriptive summary of variables from extended models for data set A and B.

Construct	Item #	Scale	M_A	SD_A	ω_A	M_B	SD_B	ω_B
Positive affect	5	1–5	3.17	0.82	0.85	3.12	0.79	0.83
Negative affect	5	1–5	1.68	0.87	0.92	1.66	0.85	0.92
Familiarity	1	1–5	3.32	1.22	NA	3.28	1.23	NA
Herding	2	1–5	3.58	0.73	0.12	3.57	0.73	0.09
Expertise	1	1–4	2.47	0.69	NA	2.45	0.69	NA
Benefits	6	1–7	3.77	1.37	0.93	3.78	1.37	0.93
Concerns	16	1–5	3.33	0.78	0.93	3.29	0.76	0.92
Negative experiences	2	1–5	2.64	0.64	0.38	2.63	0.65	0.46
Literacy	7	0–7	4.41	1.28	0.26	4.46	1.26	0.34
Vertical privacy	12	1–7	3.29	1.23	0.94	3.17	1.16	0.93
Horizontal privacy	7	1–7	3.59	1.22	0.89	3.48	1.19	0.89
Protection measures	6	1–7	3.74	1.36	0.90	3.72	1.36	0.90